Daniel C. Posmik

Contact Information

Brown University
School of Public Health
Department of Biostatistics
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Interests

I am currently working on clinically interpretable uncertainty quantification for tau pathology structures in PET imaging. Methodologically, I am interested in understanding how latent geometry and functional representation can explain how multimodal models learn.

Education

Brown University, Providence, Rhode Island USA
Ph.D. in Biostatistics, Advisor: Ani Eloyan — Exp. May 2029

University of Cincinnati, Cincinnati, Ohio USA
B.S. in Data Analytics; B.B.A. in Economics; Mathematics Minor — May 2022
A.A. in Business Administration — May 2020

Honors and Awards

McCall MacBain Finalist Award (Deferred for Ph.D.), 80 of 1,000+ applicants — 2023
Presidential Leadership Medal of Excellence, Univ. of Cincinnati, 4 yearly recipients — 2022

Work Experience

Brown University, Providence, Rhode Island USA

Graduate Student, Department of Biostatistics — Aug. 2024 — Present

- Writing a review paper on statistical methods for PET imaging, covering preprocessing, data analysis and reproducibility. (*In preparation*)
- Developing a Bayesian variance estimator for principal manifold estimation that propagates PET-to-MRI alignment uncertainty through a diffusion-map embedding, yielding posterior credible bands for the estimated brain manifold. (*In preparation*)

University of Chicago, Chicago, Illinois USA

Graduate Student at Large, Department of Statistics — Aug. 2023 — May 2024

- Research assistant for Prof. Guanglei Hong on methods for causal decomposition analysis; TA for Econometrics.

Project Associate, Energy and Environment Lab — Jan. 2023 — Aug. 2023

- Led causal analyses for a randomized controlled trial on the effect of agricultural machinery and illegal crop burning in Pakistan; analyzed large remote sensing data; contributed to grant reporting.

The Brattle Group, Boston, Massachusetts USA

Research Analyst, Electricity Practice Group — June — Dec. 2022

- Conducted econometric analyses on wholesale energy markets and electricity congestion patterns; informed a utility's $10m investment.

Publications

D.C. Posmik, Ani Eloyan (*In preparation*). Posterior Covariance Estimation on Principal Manifolds with Multiview Neuroimaging Data.

D.C. Posmik, Renaud LaJoie, Ani Eloyan (*In preparation*). A Statistical Guide to PET Data: Preprocessing, Data Analysis, and Reproducibility

D.C. Posmik (2022). Predicting International Student Enrollment by Institutional Aid: A Random and Fixed Effects Approach. *Journal of Student Financial Aid*, 51 (3), Article 4.

Skills & Languages

Programming Languages: R, Python
Machine Learning: PyTorch
Productivity: LaTeX, Git
Languages: English (fluent), German (native), French (advanced)